

AnimalWatch: Acoustic Elk Detection Device

Designed a solar-powered, MEMS-microphone-based sensor network for real-time elk detection and road safety alerting. Developed from concept through hardware prototype and pitched to academic evaluators as an investor presentation.



83/100

PITCH SCORE

6

TEAM MEMBERS

1,600

ANNUAL ELK-VEHICLE
ACCIDENTS
(NORDICS)

30-40%

PROJECTED
COLLISION
REDUCTION

Problem & Scope

ROLE	CONTEXT	TEAM	DELIVERABLES
CTO / Technical Lead	Aalto University, Product Development Project, 2024	6 members (cross-functional)	Hardware prototype, idea pitch, business plan

My contribution: As CTO, I led the technical design of the entire sensing architecture, including sensor selection and specification (MEMS microphones, DHT11), microcontroller integration (Arduino Mega),

power management design (solar-hybrid with TP4056 charge controller), and communication link budgeting (HC-12 at 433 MHz). I also co-developed the signal processing pipeline concept and presented the technical sections of the investor pitch to academic evaluators.

Across the Nordic countries, approximately 1,600 elk-vehicle collisions are recorded annually, causing fatalities, injuries, and estimated financial losses exceeding €70 million per year. Traditional wildlife monitoring methods such as camera traps and GPS collars are either expensive, invasive, or lack real-time capability. AnimalWatch addresses this gap with a non-invasive, scalable, acoustic detection system designed for remote deployment along high-risk road corridors.

Arduino Mega

INMP441 MEMS Mic

DHT11 Sensor

HC-12 433 MHz

TP4056 Charger

Solar Panel (5V 5W)

Li-Ion Battery

3D Printing

System Design

The system uses distributed MEMS microphone arrays placed at known elk migration points, combined with environmental sensors, to detect, classify, and localize elk vocalizations in real time.

Technology Stack

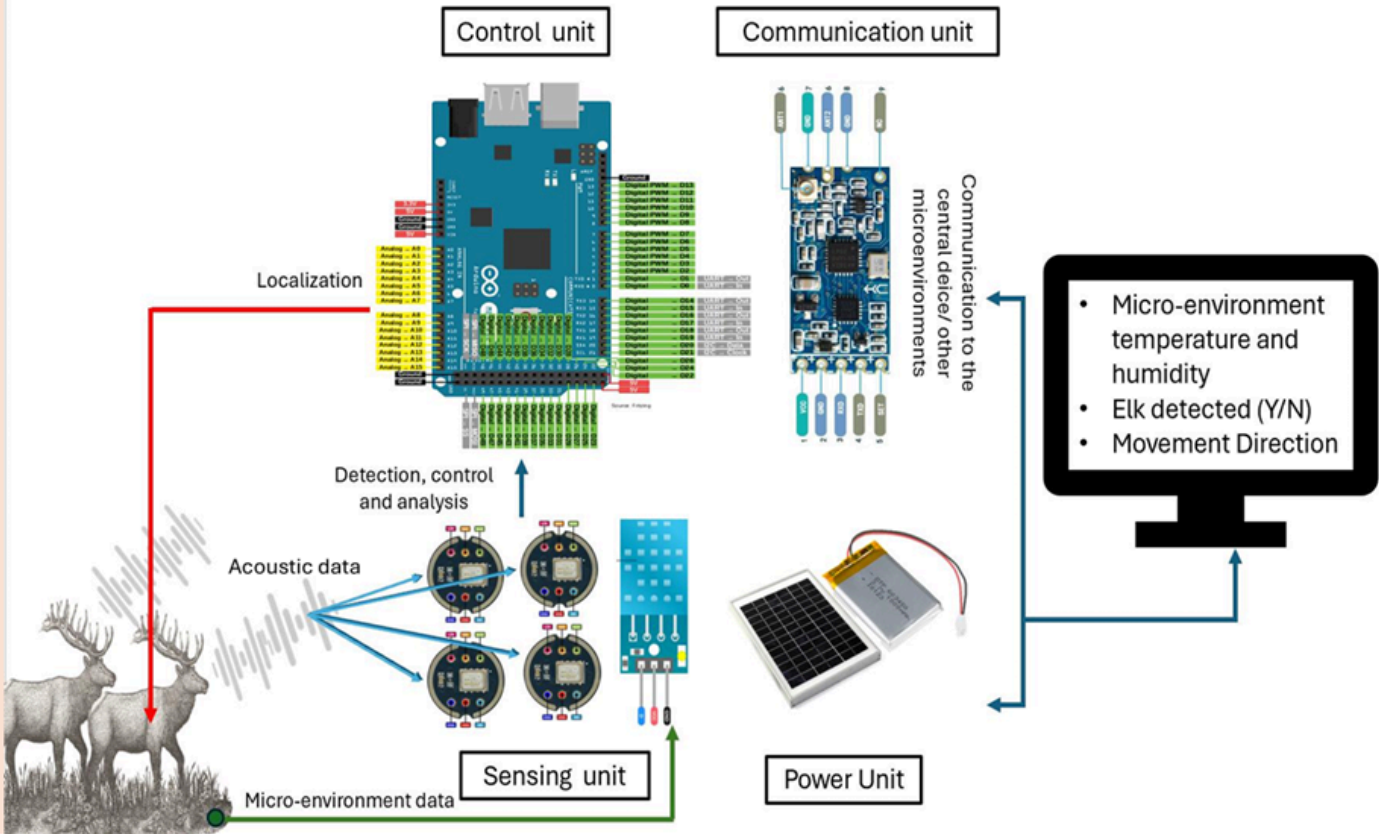


Fig 1 – Full system architecture: sensing unit (MEMS microphones + DHT11), control unit (Arduino Mega), communication (HC-12), and power management (solar + Li-Ion)

SOLUTION

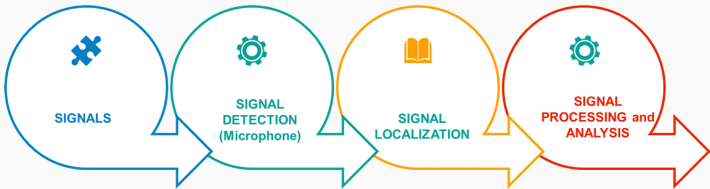


Fig 2 – Signal processing pipeline from acoustic capture through localization to movement analysis



Fig 3 – AnimalWatch team during a project work session at Aalto University

ENGINEERING DECISION: SENSING APPROACH

CONSIDERED

CHOSEN

GPS/satellite collars (precise but invasive, high per-unit cost)	MEMS microphone array with environmental sensors
Camera traps (visual but limited range, high maintenance)	
Radar-based detection (long range but expensive, overkill for species ID)	
Rationale: The INMP441 MEMS microphone offers a 20 Hz to 20 kHz frequency response with 64-70 dB SNR at an operating voltage of just 1.6-3.6V, making it ideal for capturing elk vocalizations while keeping power consumption minimal. Combined with the DHT11 for correlating elk activity with temperature and humidity patterns, the acoustic approach is non-invasive, scalable across terrain, and far cheaper per monitoring point than GPS collars or radar. An array of four microphones per node enables time-delay-based localization without physical contact with the animals.	

ENGINEERING DECISION: COMMUNICATION MODULE	
CONSIDERED	CHOSEN
WiFi (short range, high power in remote forest)	HC-12 transceiver at 433 MHz (1 km range)
LoRa (long range but higher cost and complexity)	
Cellular/GSM (coverage gaps in remote areas)	
Rationale: The HC-12 provides up to 1 km line-of-sight range at 20 dBm transmit power while consuming only 16 mA in TX and 3.5 mA in RX mode, making it well-suited for a solar-powered device in forested terrain. Its configurable baud rate (1,200-115,200 bps) handles the modest data throughput of acoustic classification results and environmental readings. The 433 MHz ISM band requires no licensing in the EU. For a prototype validating the detection concept, this represented the best trade-off between range, power, and simplicity before scaling to LoRaWAN in a production design.	

Market Positioning

The wildlife tracking system market is growing at a CAGR of 14.3%, projected to reach \$42.8 billion by 2031. AnimalWatch

differentiates on cost-effectiveness and environmental integration.

FEATURE	NAVTECH RADAR	INFRARED CAMERA	WARNING SIGN SYSTEMS	ANIMALWATCH
Real-time detection	✓	✓	✓	✓
Precise localization	✓	✓	✗	✓
Cost-effective	✗	✗	✓	✓
Non-invasive	✓	✓	✓	✓
Environmental integration	✗	✗	✗	✓
Multi-species adaptable	✓	✓	✓	✓

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PITCH EVALUATION

Evaluator Scores

The idea pitch was evaluated by academic reviewers across eight criteria. The project scored 83/100 overall, with the strongest marks in technical content (14/15) and soundness of solution (13/15).

13/15 PROBLEM CLARITY	13/15 SOLUTION SOUNDNESS	14/15 TECHNICAL CONTENT	7/10 USE OF RESOURCES
8/10 COURSE METHODS	11.5/15 STYLE CONSISTENCY	8.5/10 OVERALL PRESENTATION	8/10 PERSONAL IMPRESSION

What Made This Hard

Power budget in remote forest

Designed a solar-hybrid power system with TP4056 charge controller, Li-Ion battery, and aggressive duty-cycling. Acoustic sampling runs in short bursts with deep sleep between cycles to keep average consumption below the solar harvest rate during Nordic winter daylight hours.

Elk vocalization classification

Elk calls overlap spectrally with ambient forest noise (wind, other fauna). The prototype pipeline uses band-pass filtering and a threshold-based classifier; the evaluators rightly noted that a trained ML model is needed for production accuracy. We scoped the prototype to prove detection feasibility, not production-grade classification.

Enclosure and weatherproofing

The evaluators flagged that the presentation did not address enclosure design, which is critical for a device deployed year-round in Nordic conditions. The prototype housing was 3D-printed for lab demonstration; an IP65-rated enclosure with acoustic venting is required for field deployment and was identified as a key next step.

Presentation style consistency

With six team members presenting, evaluators noted inconsistency in animation use, audio quality, and slide density between presenters. This was the lowest-scoring criterion (11.5/15) and a direct lesson in coordinating team deliverables for a unified pitch.

What I Learned

REFLECTION

This was my first project that combined hardware prototyping with a full investor-pitch format, bridging engineering design with business justification. Leading the technical architecture taught me to defend component-level decisions not just on specs, but on power, cost, and deployment trade-offs that matter to non-technical stakeholders. The evaluator feedback on missing enclosure discussion and citation

gaps was a strong lesson: a pitch audience fills gaps with doubt, so every foreseeable question needs to be preempted.

NEXT STEPS (IF CONTINUED)

Replace the threshold-based classifier with a trained ML model (e.g., a lightweight CNN on Mel spectrograms) running on a low-power MCU such as ESP32-S3 with edge inference. Design an IP65-rated weatherproof enclosure with acoustic venting for year-round Nordic deployment. Migrate the communication stack from HC-12 to LoRaWAN for multi-kilometer mesh networking. Conduct field trials in collaboration with Metsähallitus (Finnish Forest Administration) to validate detection accuracy in real habitat conditions.